
Parasolid V35.0

What's New in Parasolid

July 2022

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1.1 Introduction

This document describes functional enhancements and interface changes that have been added to Parasolid V35.0, and gives information on any other matters that are specific to this release.

1.2 Lattices

This section describes improvements that have been made to Parasolid's support for representing lattices in Parasolid V35.0. Full details on Parasolid lattices can be found in Chapter 20, "Lattice Geometry", of the Parasolid *Functional Description*.

1.2.1 Creating a patterned rod and ball lattice

In Parasolid V35.0 you can now create a complete lattice by applying a pattern to an initial portion of the lattice. You can optionally supply a user-defined callback function that can deform the pattern.

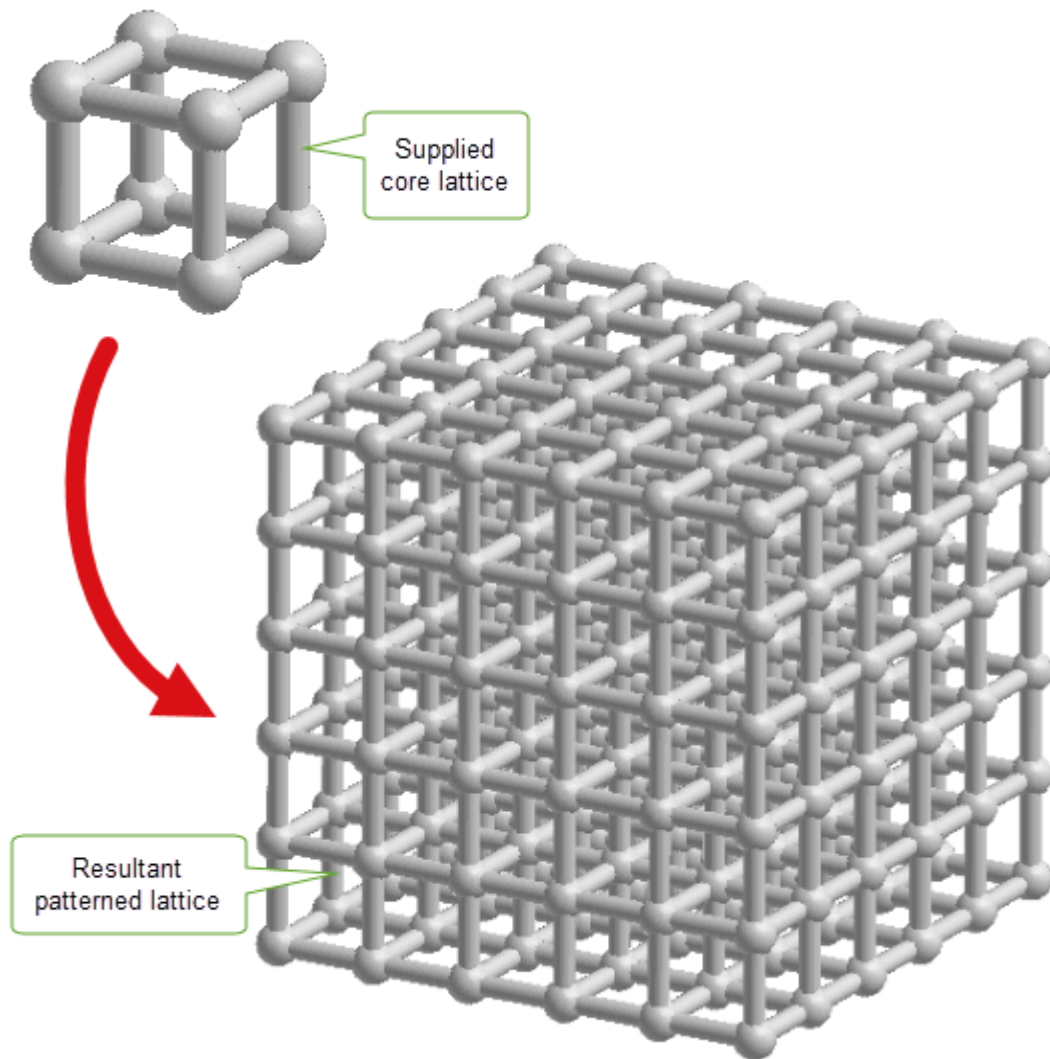


Figure 1-1 Creating a patterned lattice using a supplied lattice

For more information, see Section 20.6, “Creating a lattice structure from a core lattice” in the *Parasolid Functional Description*.

1.2.1.1 Interface changes

New	PK_LATTICE_make_patterned, PK_LATTICE_make_patterned_o_t, PK_LATTICE_make_patterned_r_f, PK_LATTICE_make_patterned_r_t
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1.2.2 Creating a lattice-like body from a mesh body

In Parasolid V35.0, you can pattern a facet body into a lattice-like complex mesh body. This is achieved by transforming instances of the facet body and deforming the result using a user-defined callback function to match a rectilinear pattern.

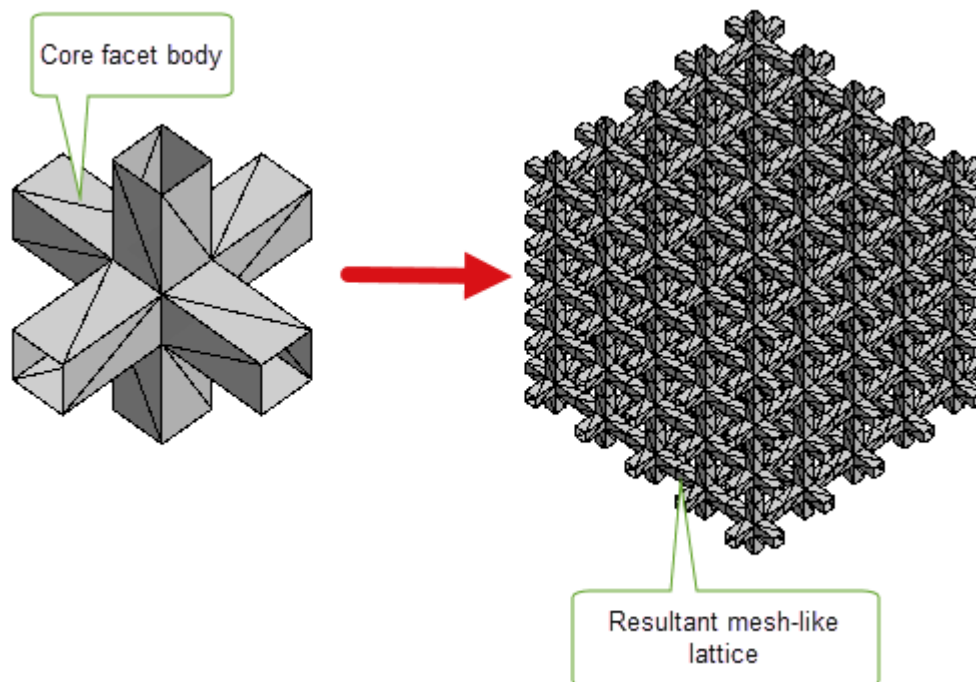


Figure 1–2 Creating a lattice-like body from a mesh body

For more information, see Section 20.7, “Creating a lattice-like mesh body” in the Parasolid *Functional Description*.

1.2.2.1 Interface changes

New	PK_BODY_make_patterned, PK_BODY_make_patterned_o_t, PK_BODY_make_patterned_r_f, PK_BODY_make_patterned_r_t, PK_BODY_patterned_t, PK_pattern_bound_t, PK_pattern_cell_t, PK_SKEWBOX_sf_t, PK_pattern_t, PK_pattern_type_t, PK_pattern_rectilinear_o_t,
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1.2.3 Creating blended lattices

In V35.0, you can now create blended lattices where the blend size of the ltopologies is the same throughout the lattice or where the blend size varies. Blending a lattice allows you to increase the stiffness of the lattice thereby improving its structural strength.

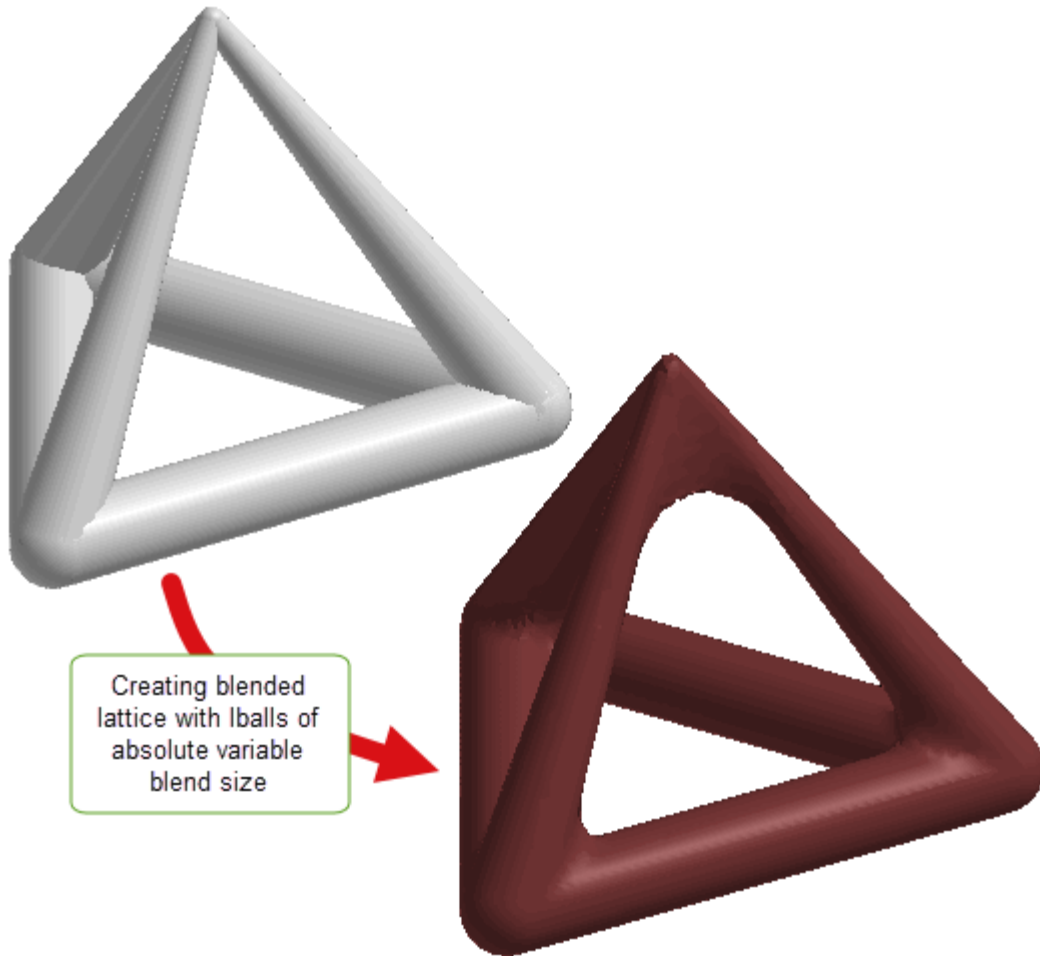


Figure 1-3 Blended lattice with ltopologies of different blend sizes

For more information, see Chapter 20.5, “Creating a blended lattice” in the *Parasolid Functional Description*.

Note: This functionality is not complete in the current release and the description above is included for information only. We expect full functionality to be available in a future release of Parasolid though you should note that the information here is subject to change.

1.2.3.1 Interface changes

Changed	PK_LATTICE_create_by_graph: new options; lball_blend_type and lball_blend_size PK_LATTICE_graph_cyl_o_t, PK_LATTICE_graph_cone_o_t: new option lball_blend_sizes
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1.2.4 Merging coincident ltopologies when clipping a lattice

You can choose whether coincident ltopologies that have been created as a result of clipping should be merged. This allows a cleaner clipping result.

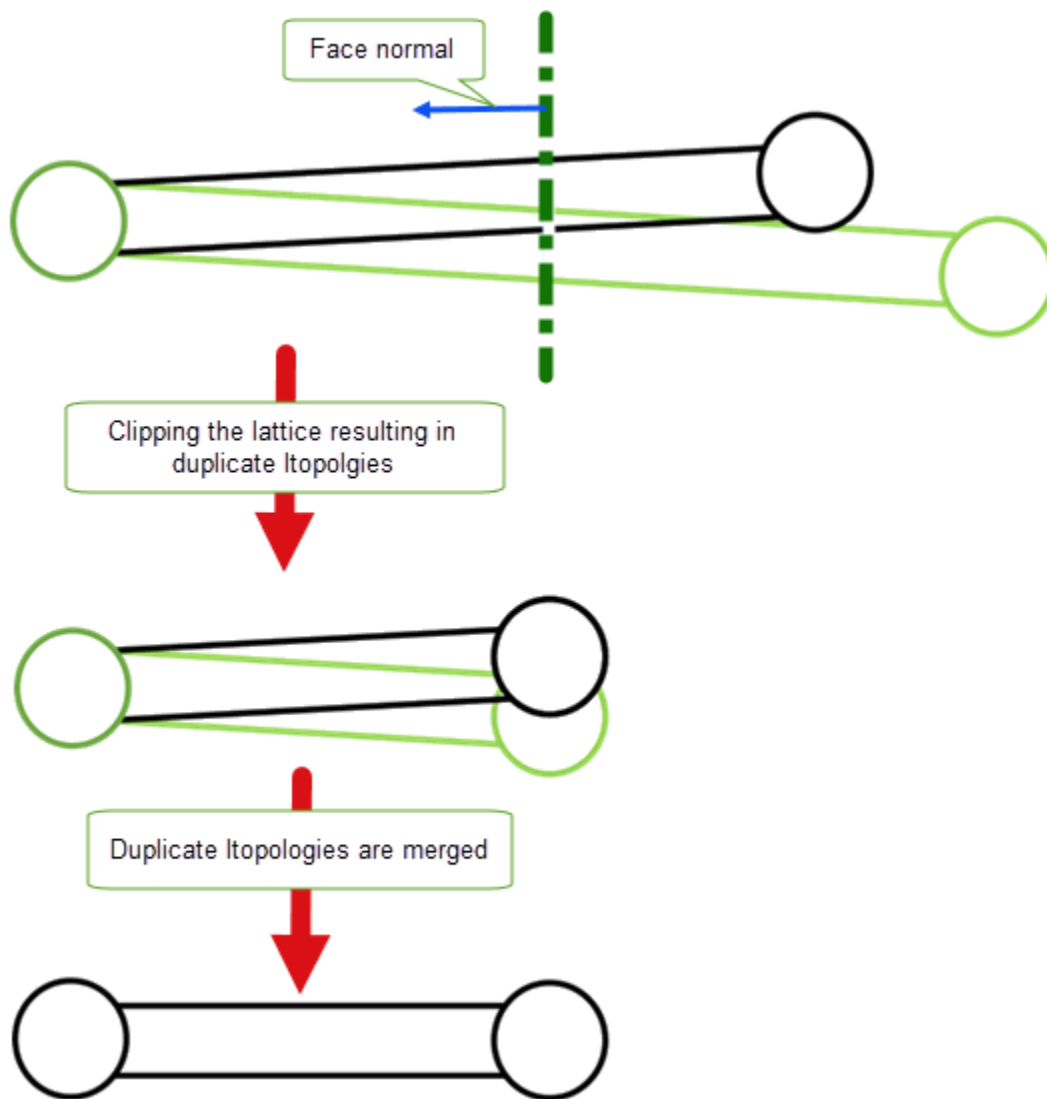


Figure 1–4 Merging duplicate Itopologies when clipping a lattice

For more information, see Section 55.6.4.2, “Merging duplicate Itopologies when clipping lattices” in the *Parasolid Functional Description*.

1.2.4.1 Interface changes

New	PK_LTOPOL_dup_merge_t: used by merge_duplicates
Changed	PK_LATTICE_clip: new options; merge_duplicates and duplicate_tolerance

1.2.5 Finding the non-axis aligned box of a lattice

Parasolid V35.0 allows you to create a tight non-axis aligned box for a given lattice.

For more information, see Section 26.6.1.1, “Axis-aligned and non axis-aligned boxes” in the Parasolid *Functional Description*.

1.2.5.1 Interface changes

New	PK_LATTICE_find_nabox
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1.3 Model structure

This section describes improvements that have been made to Parasolid’s model structure for Parasolid V35.0. Full details on Parasolid’s model structure are given in Volume 3: Basic Concepts of the Parasolid *Functional Description*.

1.3.1 Rendering models outside the size box

Further improvements have been made in our support for representing models outside the size box. Hidden line rendering now supports a collection of bodies that are being instanced to positions outside of the size box.

Note: This functionality is not complete in the current release and the description above is included for information only. We expect full functionality to be available in a future release of Parasolid though you should note that the information here is subject to change.

1.3.1.1 Interface changes

None

1.3.2 Support for models of different scales

Parasolid V35.0 provides support for models whose parts are of different scales by allowing geometry created at a larger scale to be scaled and trimmed so they can interact with entities at a smaller scale. This functionality can be useful when working with assemblies that consist of mixed scale parts.

For more information, see Section 14.9, “Representing models whose parts are of different scales” in the Parasolid *Functional Description*.

In addition, you can also generate facets for a set of models which are created at different scales

For more information, see Section 107.2.22, “Facet generation on bodies created at different scales” in the Parasolid *Functional Description*.

1.3.2.1 Interface changes

New	PK_BODY_enlarge, PK_BODY_enlarge_o_t, PK_BODY_enlarge_result_t, PK_BODY_enlarge_r_t, PK_BODY_enlarge_r_f PK_GEOM_enlarge, PK_GEOM_enlarge_o_t, PK_GEOM_enlarge_result_t, PK_GEOM_enlarge_r_t, PK_GEOM_enlarge_r_f PK_scale_factor_t: used by scale_factors SDL/TYSA_SCALE_FACTOR
Changed	PK_TOPOL_facet_mesh_o_t and PK_TOPOL_facet_mesh_2_o_t: new options; bodies_with_scales, scale_factors, n_bodies_with_scales

1.3.3 Labelling the scale of a part

A new system attribute has been added to allow your application to label the scale of a part.
For more information on the new system attribute added, see Appendix A, “System Attribute Definitions” in the Parasolid *Functional Description*.

1.3.3.1 Interface changes

New	SDL/TYSA_SCALE_FACTOR
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1.4

Application support

This section describes improvements that have been made to Parasolid’s application support functionality for Parasolid V35.0. Full details on Parasolid’s support for application development are given in Volume 2 Application Development Essentials and Volume 14 Application Support of the Parasolid *Functional Description*.

1.4.1 Expression evaluators

Parasolid V35.0 allows an application to supply a mathematical formula to be evaluated instead of supplying an evaluator function. For more information on expressions, see Chapter 24, “Expressions”.

For information on how you can use an expression evaluator to describe how an input face can be deformed instead of using a callback function, see Section 72.4.8, “Deformation options” in the Parasolid *Functional Description*.

1.4.1.1 Interface changes

New	PK_EXPR_sf_t; used by eval_expr PK_EXPR_value_type_t, PK_EXPR_largest_value_t, PK_EXPR_value_t
Changed	PK_FACE_change_data_deform: new option; eval_expr

1.4.2 Partition creation and enquiries

In Parasolid V35.0, new functions have been added to create and enquire partitions.

1.4.2.1 Interface changes

New	PK_PARTITION_create, PK_PARTITION_create_o_t, PK_PARTITION_create_r_t, PK_PARTITION_create_r_f PK_PARTITION_ask_pmarks_2, PK_PARTITION_ask_pmarks_2_o_t, PK_PARTITION_ask_pmarks_2_r_t, PK_PARTITION_ask_pmarks_2_r_f PK_PARTITION_ask, PK_PARTITION_ask_o_t, PK_PARTITION_ask_r_t, PK_PARTITION_ask_r_f
Changed	PK_PARTITION_create_empty and PK_PARTITION_ask_pmarks have been deprecated

1.5 Sheet modelling

This section describes improvements that have been made to Parasolid's sheet modelling functionality for Parasolid V35.0. Full details on Parasolid's sheet modelling functionality are given in Volume 7: *Working with Sheets and Wires*, of the *Parasolid Functional Description*.

1.5.1 Creating neutral sheets from disconnected walls

In Parasolid V35.0, you can create neutral sheets from walls consisting of disjoint offsets of the same surface.

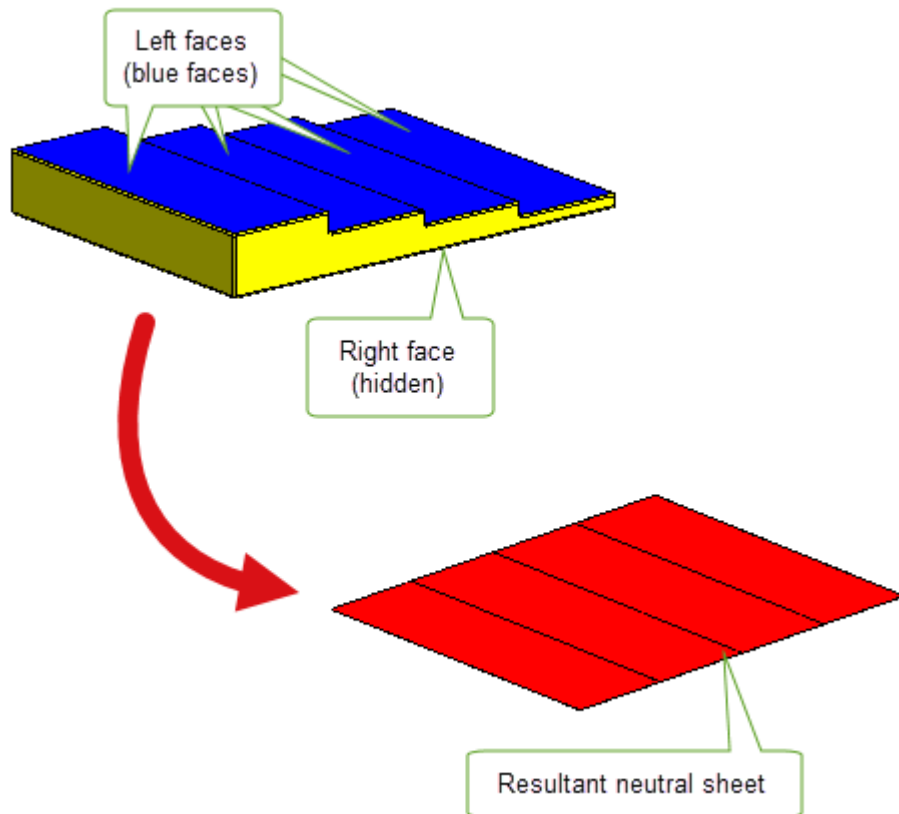


Figure 1–5 Neutral sheet creation from disconnected walls

1.5.1.1 Interface changes

None, but changes affect PK_FACE_make_neutral_sheet_2

1.5.2 Returning tracking information when sectioning with sheet

Parasolid V35.0 allows you to return topological tracking information when sectioning a body with a sheet.

For more information, see Section 55.4, “Options for global and local sectioning”, in the *Parasolid Functional Description*.

1.5.2.1 Interface changes

Changed	PK_BODY_section_with_sheet_2: new option; tracking
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1.5.3 Creating sheet bodies with touching loops

In Parasolid V35.0, you can create planar sheet bodies that contain touching loops from arrays of positions.

1.5.3.1 Interface changes

None but changes affect PK_BODY_create_sheet_planar

1.6 Model interrogation

This section describes improvements that have been made to Parasolid's model interrogation functionality for Parasolid V35.0. Full details on Parasolid's interrogation functionality are given in Volume 4: Model Interrogation, of the Parasolid *Functional Description*.

1.6.1 Checking validity of assemblies

Parasolid V35.0 allows you to perform checks on assemblies and return relevant fault information. This functionality will enable you to produce better quality models.

For more information, see Chapter 30, "Checking" in the Parasolid *Functional Description*.

1.6.1.1 Interface changes

New	PK_ASSEMBLY_check, PK_ASSEMBLY_check_o_t, PK_ASSEMBLY_check_r_t, PK_ASSEMBLY_check_r_f
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1.7 Model editing

This section describes improvements that have been made to Parasolid's model editing functionality for Parasolid V35.0. Full details on Parasolid's editing functionality are given in Volume 9: Offsetting Operations, and Volume 10: Model Editing, of the Parasolid *Functional Description*.

1.7.1 Improved error reporting in delete blends

In V35.0, blend faces that failed to be deleted will now be reported in the Parasolid Report.

For more information, see Section 63.6, "Deleting blend faces" in the Parasolid *Functional Description*.

1.7.1.1 Interface changes

None, but changes affect PK_FACE_delete_blends

1.7.2 Improved support when healing non-manifold wounds

In Parasolid V35.0, you can now delete and heal non-manifold wounds

1.7.2.1 Interface changes

None, but changes affect PK_FACE_delete_2

1.8 Outlining

This section describes improvements that have been made to Parasolid's outlining functionality for Parasolid V35.0. Full details on Parasolid outlining is given in Volume 6: Profiling and Surfacing, of the Parasolid *Functional Description*.

1.8.1 Performance improvements when creating outline curves

Parasolid V35.0, has improved the performance of curve outline functionality. Parasolid can use the outline of the body boxes, to provide a better estimate of the view-size of the body. This is beneficial when working with assemblies containing parts of varying sizes and distances from the eyepoint.

1.8.1.1 Interface changes

None, but changes affect PK_BODY_make_curves_outline and PK_BODY_make_persp_outline

1.9 Booleans

This section describes improvements that have been made to Parasolid's Boolean functionality for Parasolid V35.0. Full details on Parasolid Booleans are given in Volume 8: Imprinting and Booleans, of the Parasolid *Functional Description*.

1.9.1 Repairing non-manifold edges in resulting bodies of mixed geometry

In Parasolid V35.0, you can attempt to repair non-manifold edges that have been created as a result of a boolean operation on bodies of mixed geometry.

Note: This functionality is not complete in the current release and the description above is included for information only. We expect full functionality to be available in a future release of Parasolid though you should note that the information here is subject to change.

1.9.1.1 Interface changes

None, but changes affect PK_BODY_boolean

1.10 Faceting and rendering

This section describes improvements that have been made to Parasolid's faceting and rendering functionality for Parasolid V35.0. Full details on Parasolid's faceting and rendering functionality are given in Volume 15: Graphics Support, of the Parasolid *Functional Description*

1.10.1 Improved control when rendering parametric hatch lines

In Parasolid V35.0 you can now specify the exact number of parametric hatch lines to draw when rendering a model. A new system attribute has been added to allow your application to indicate that parametric hatch lines should be drawn.

For more information, see Section 105.3.18.3, "Parametric hatching" in the Parasolid *Functional Description*.

For more information on the new system attribute added, see Appendix A.3.5.3, "Parametric hatching (numeric control)" in the Parasolid *Functional Description*.

1.10.1.1 Interface changes

Changed	PK_TOPOL_render_line: new options; param_u_number and param_v_number PK_render_param_t; new value: pk_render_param_number_free_c
New	SDL/TYSA_PARAM_NUM_HATCH

1.10.2 Using viewports to limiting the number of facets returned

In Parasolid V35.0 you can choose to return only facets that lie on faces which overlap with at least one supplied viewport.

For more information, see Section 107.2.23, "Using viewports to limit the number of facets returned", in the Parasolid *Functional Description*.

1.10.2.1 Interface changes

Changed	PK_TOPOL_facet_2, PK_TOPOL_render_facet: new options n_viewports and viewports
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1.10.3 Performance improvements for hidden line overlap

Parasolid V35.0, has improved the performance of the overlap option in hidden line and it can now be used when rendering concurrently

1.10.3.1 Interface changes

None, but changes affect PK_TOPOLOGY_render_line

1.11 New system attributes

This section lists the new system attributes that have been added to Parasolid V35.0.

See Appendix A, "System Attribute Definitions", in the Parasolid *Functional Description* for more information.

- SDL/TYSA_SCALE_FACTOR
- SDL/TYSA_PARAM_NUM_HATCH

1.12 New session switches

This section lists the new session switches that have been added to Parasolid V35.0.

Session switches should only be used when updating models created using an older version of Parasolid (including all versions prior to V28.1), and only when directed to do so by Parasolid Technical Support. These switches are not required when performing new modelling operations.

See Section 116.3, "Reproducing behaviour using the original system", in the Parasolid *Functional Description* for more information.

There are no new session switches in this release.

1.13 Newly deprecated interfaces and tokens

This section lists the interfaces and tokens that have been deprecated in Parasolid V35.0. For a list of all deprecated functions, see the PK Deprecated Functions list in the Parasolid PK Reference.

- PK_PARTITION_create_empty
- PK_PARTITION_ask_pmarks

1.14 Forthcoming functionality

The functionality below is blocked in the current release. We expect to unblock it in a future release of Parasolid.

- Section 1.2.3, "Creating blended lattices"
- Section 1.3.1, "Rendering models outside the size box"
- Section 1.9.1, "Repairing non-manifold edges in resulting bodies of mixed geometry"

1.15 Changes to the Parasolid documentation

In addition to documentation covering the changes and additions to Parasolid functionality described elsewhere in this chapter, the following changes have been made to the Parasolid documentation suite:

- The documentation for face blending has been updated to clarify restrictions when trimming a face blend. For more information, see Section 77.6.2, “Trimming blend walls and returning the body” in the Parasolid *Functional Description*.

1.16 Platforms

1.16.1 Changes for Parasolid V35.0

This section describes changes to the platforms on which Parasolid ships that are relevant to the V35.0 release. For full details of changes to platforms, please see the customer letter accompanying the Parasolid V35.0 release.

- We have changed the C++ code in Parasolid to being compiled as c++17 on all platforms.
- The GCC 4.8 build of the Intel Linux platform has been dropped. The GCC 7.3 build is the only build on this platform.
- We have replaced the Visual Studio 2019 build of ARM Win with a Visual Studio 2022 build
- The ARM Android builds are now compatible with Android 9 and later and are built with NDK21b.

1.16.2 Plans for Parasolid V35.1

This section describes our current plans for Parasolid V35.1. For more details, please see the customer letter accompanying the Parasolid V35.1 release.

Parasolid V35.1 is expected to ship in the first quarter of 2023

We intend to replace:

- The NDK21b build of ARM Android with one built on NDK23b, still compatible with Android 9 onwards.
- The ARM MacOS and ARM iOS builds, built on macOS 11, with new builds on macOS 13, supporting macOS 11 and later and iPadOS 15 or later.
- The Intel MacOS and Intel iOS builds, built on macOS 11 and supporting iPadOS 13 and later, with new builds on macOS 13 and supporting macOS 11 or later and iPadOS 15 and later. These builds are intended to set the final build standard for these platforms; we do not intend to update them again before they are dropped. Apple plans to stop selling Intel-based computers, so at some point, it will be necessary for us to give up building software for them. The date for this is not yet set.

1.17 New PK interface tokens

Information to come.

PKPK_BODY_patterned_t

PK_BODY_patterned_success_c	26990
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PK_check_returns_t

PK_check_returns_5_c	23666
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PK_check_state_t

PK_ASSEMBLY_state_invalid_id_c	26773
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PK_EXPR_value_type_t

PK_EXPR_value_type_integer_c	27030
PK_EXPR_value_type_real_c	27031
PK_EXPR_value_type_vector_c	27032
PK_EXPR_value_type_entity_c	27033

PK_GEOM_enlarge_t

PK_GEOM_enlarge_exact_c	26970
PK_GEOM_enlarge_outside_c	26971
PK_GEOM_enlarge_convert_c	26972
PK_GEOM_enlarge_approx_c	26973
PK_GEOM_enlarge_failed_c	26974

PK_LATTICE_patterned_t

PK_LATTICE_patterned_success_c	27010
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PK_LBALL_blend_t

PK_LBALL_blend_none_c	27040
PK_LBALL_blend_absolute_c	27041
PK_LBALL_blend_relative_c	27042

PK_LTOPOL_dup_merge_t

PK_LTOPOL_dup_merge_no_c	27020
PK_LTOPOL_dup_merge_yes_c	27021

PK_pattern_cb_status_t

PK_pattern_cb_status_ok_c	27060
PK_pattern_cb_status_fail_c	27061

PK_pattern_type_t

PK_pattern_type_rectilinear_c	26980
PK_pattern_type_original_c	26981

PK_REPORT_3_t

PK_REPORT_3_render_lattice_c	26842
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PK_render_param_t

PK_render_param_spaced_c	20182
PK_render_param_yes_c	PK_render_param_spaced_c
PK_render_param_spaced_free_c	20183
PK_render_param_free_c	PK_render_param_spaced_free_c
PK_render_param_number_free_c	20184

PK_scale_factor_t

PK_scale_factor_1000x_c	26963
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PK_track_faces_t

PK_track_faces_yes_c	27050
PK_track_faces_no_c	27051

1.18 New PK interface error codes

PK_ERROR_code_t

PK_ERROR_lattice_geometry	5277
PK_ERROR_pattern_failed	5278
PK_ERROR_vectors_coplanar	5279
PK_ERROR_partial_pmark	5280
PK_ERROR_too_many_facets	5281
PK_ERROR_basis_not_right_handed	5282
PK_ERROR_bad_expr	5283

PK_ERROR_expr_undef_symb	5284
PK_ERROR_expr_duplicate_def	5285
PK_ERROR_outside_box	5286